2.

1. Electron rest mass m_0, photon frequency ν; assume it absorbs the photon completely and moves at v; momentum conservation then gives

$$\frac{h\nu}{c} = \frac{m_0 v}{\sqrt{1-(v^2/c^2)}} \tag{1}$$

The total energy after collision is

$$E = mc^2 = \frac{m_0 c^2}{\sqrt{1-(v^2/c^2)}} \tag{2}$$

Eliminate v using (1)、(2):

$$E = \sqrt{m_0^2 c^4 + h^2\nu^2} \tag{3}$$

Electron - photon total energy before collision:

$$E_0 = h\nu + m_0 c^2 \tag{4}$$

Using (3)、(4),

$$E^2 - E_0^2 = (m_0^2 c^4 + h^2\nu^2) - (h\nu + m_0 c^2)^2 =$$

$$-2h\nu m_0 c^2 \neq 0 \tag{5}$$

This violates energy conservation, so the supposed process cannot occur.

2. Metal-bound electron + incident photon: momentum is not conserved in photoemission. Use energy conversion.

Escape from metal needs at least work W (work function). After escape: speed v, photon energy $h\nu$, electron rest mass m_0. Photoemission requires

$$h\nu \geqslant \left[\frac{m_0 c^2}{\sqrt{1-(v^2/c^2)}} - m_0 c^2\right] + W \tag{6}$$

The emitted speed v is much less than light speed; hence

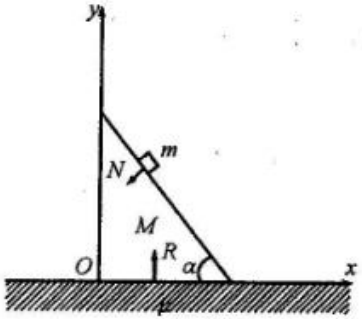
$$\frac{1}{\sqrt{1-(v^2/c^2)}} \approx 1 + \frac{1}{2}\frac{v^2}{c^2} + \cdots \tag{7}$$

Drop higher terms; substitute the first two of (7) in (6):

$$h\nu \geqslant \frac{1}{2} m_0 v^2 + W \tag{8}$$

Thus $h\nu \geqslant W$ suffices for (8) and photoemission.

3. For ease, consider the small block on BC. Oxy as shown; a_x、a_y are x, y acceleration magnitudes. N, R: block - wedge and wedge - table normal forces. Let a_M denote the wedge acceleration magnitude. Block equations:



$$ma_x = N\sin\alpha \tag{1}$$

$$ma_y = mg - N\cos\alpha \tag{2}$$

Wedge equations:

$$Ma_M = N\sin\alpha - \mu R \tag{3}$$

$$R = N\cos\alpha + Mg \tag{4}$$

The block is constrained to the slope: